Ansible Architecture ...

- **Host Inventory**: Determines the target machines where the tasks will be executed.
  - **Default Path**: /etc/ansible/hosts
  - **Examples:**
    - **Host Ranges:**

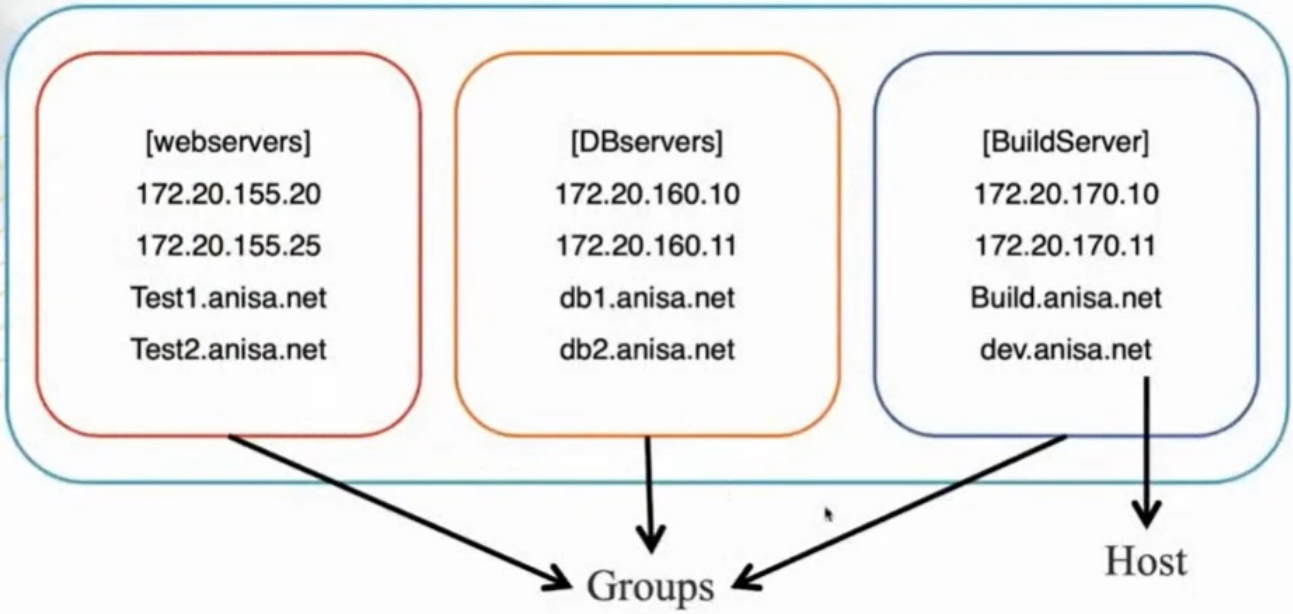
```
test[40:50].anisa.net
```

    - **Groups of Groups:**

```
[AnisaServers:children]
build-servers
webservers
```




Host Inventory





Ansible Architecture ...

- **Host Inventory**: Determines the target machines where the tasks will be executed.
  - **Default Path**: /etc/ansible/hosts
  - **Examples:**
    - **Host Ranges:**

```
test[40:50].anisa.net
```

اگه بخوایم range بسازیم

که host 11 تعریف کنه

    - **Groups of Groups:**

```
[AnisaServers:children]
build-servers
webservers
```

گروهی از گروه‌ها



Ansible Architecture ...

- **Host Inventory**: Determines the target machines where the tasks will be executed.
  - **Default Path**: /etc/ansible/hosts

یه فایل که برای Ansible مشخص میکنه که کجا

کارهاشو انجام بده

  - **Example:**
    - **Ungrouped hosts:**

```
build.anisa.net
172.20.x.x
Myserver=172.20.1.2
```

یه تا host

    - **A collection of hosts:**

```
[webservers]
test1.anisa.net
test2.anisa.net
172.20.x.x
```

چندتا host



Ansible Architecture ...

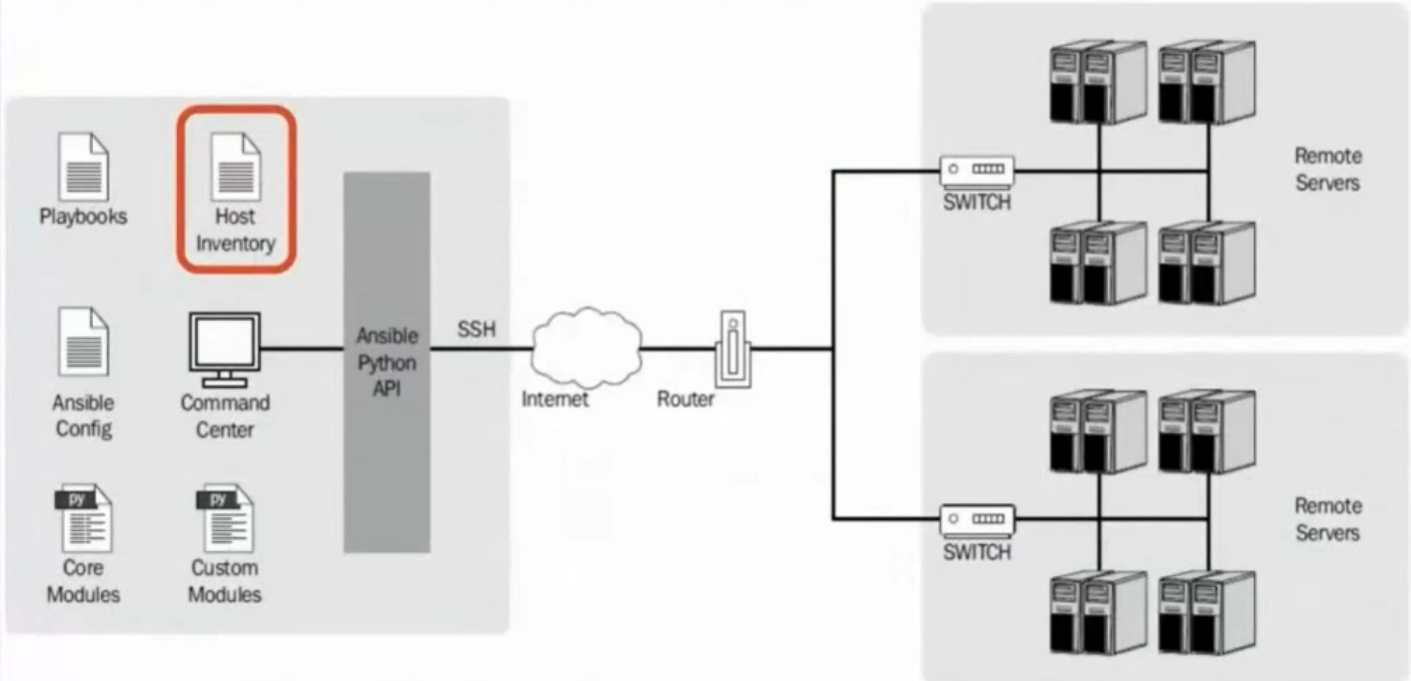
- **Host**: A host is simply a remote machine that Ansible manages.
  - **Example:**

DNS → build.anisa.net or 172.20.x.x

- **Group**: A group consists of several hosts assigned to a pool that can be conveniently targeted together.
  - **Example:**

```
[build-servers]
build.anisa.net
build-dev.anisa.net
172.20.x.x
172.20.x.x
```


Ansible Architecture ...



Ansible Vs Puppet Vs Chef Vs SaltStack

	Ansible	Chef	Puppet	SaltStack
Programming language	Python	Ruby, Erlang	C++, Clojure	Python
Configuration language	YAML, JSON	Ruby	Proprietary	YAML
Database	n/a	PostgreSQL	PuppetDB	n/a
Transport	ssh	RabbitMQ	Mcollective	ZeroMQ
Deployment method	push	pull	pull	push
Master	A, B, L, O, S	Linux	Linux	L, B
Agent	n/a	A, B, L, O, S, W	A, B, L, O, S, W	A, B, L, O, S, W
Agentless	yes	no	no	yes
Public Cloud	AM	AM/AZ/PR	no	no
Cloud management	AM, AZ, OS, GCP	Fog driver	AM, AZ, VM, GCP	Salt Cloud
Architecture	server	client, server	client, server	client, server
Step-by-step	yes	yes	no	no
Enterprise GUI	Ansible Tower	Opscode Manage	Puppet Enterprise	SaltStack Enterprise
Opensource GUI	Semaphore	Chef Manage	Foreman	Saltpad, Saltshaker
Github Stars	25,283	4,979	4,628	8,080
Github Forks	8,790	2,074	1,901	3,762
Contributors	2,957	519	455	1,885
Commits	32,760	19,692	25,816	85,485
Enterprise version	yes	yes	yes	yes

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Ansible Vs Puppet Vs Chef Vs SaltStack

Nov 26,2019

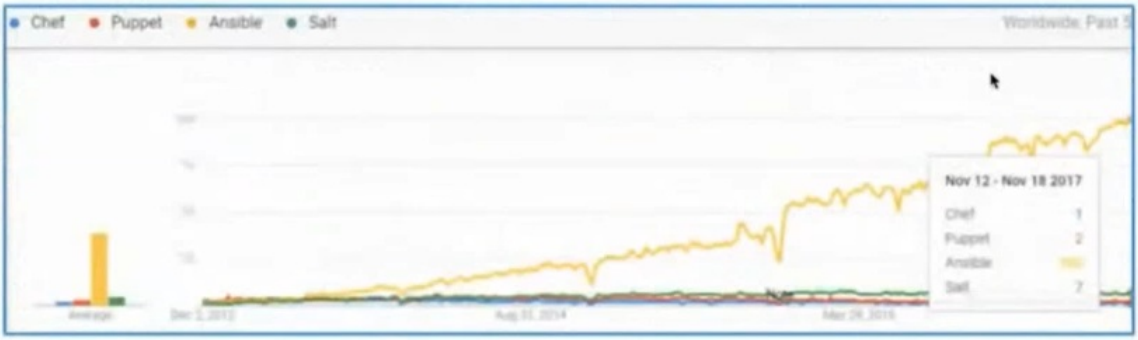
Metrics	Chef	Puppet	Ansible	Saltstack
Availability	✔	✔	✔	✔
Ease of Setup	Not very easy	Not very easy	Easy	Not very easy
Management	Not very easy	Not very easy	Easy	Easy
Scalability	Highly Scalable	Highly Scalable	Highly Scalable	Highly Scalable
Configuration language	DSL(Ruby)	DSL(PuppetDSL)	YAML(Python)	YAML(Python)
Interoperability	High	High	High	High
Pricing (upto 100 nodes)	$13700	$11200-$19900	$10,000	$15,000(approx.)

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Ansible Vs Puppet Vs Chef Vs SaltStack

popularity of the tools



Ansible Vs Puppet Vs Chef Vs SaltStack

- Conf. Lang.
- GitHub Activity
- Enterprise Cost
- Popularity
- Success Story

puppet

NYSE ICE

"75% of ICE's(Intercontinental Exchange) 20,000 servers are managed by Puppet Enterprise. They increased from 300 servers per admin to 700 servers per admin. Provisioning dev environments reduced from 1 or 2 days to just 21 minutes."

Reference: www.puppet.com/resources/customer-stories/nyse

LinkedIn SALTSTACK

"LinkedIn had about 5,000 Salt Minions under management 4 years ago. That number has ballooned to more than 70,000 today to manage their infrastructure."

Reference: www.saltstack.com/saltstack-at-web-scale-better-stronger-faster/

FATMAP

"Using Ansible, we've been able to cut down certain processes from 17 hours to 3 minutes. Ansible is unquantifiable in its benefit."

Reference: www.ansible.com/success-stories/fatmap

ANSIBLE

GANNETT

"Chef has increased the effectiveness and the speed of our development cycle. 30% of Gannett's technology organization used Chef but Gannett has aggressive plans for 100% adoption this year 2016."

Reference: www.chef.io/customers/gannett/

CHEF



Ansible Vs Puppet Vs Chef Vs SaltStack

- Conf. Lang.
- GitHub Activity
- Enterprise Cost
- Popularity
- Success Story



- ✓ Puppet and Chef are old players, puppet has wider adoption
- ✓ SaltStack and Ansible are new players, and Ansible looks very promising with the growing trend



Ansible Vs Puppet Vs Chef Vs SaltStack

- Conf. Lang.
- GitHub Activity
- Enterprise Cost
- Popularity
- Success Story

puppet

Puppet Enterprise : $12,000

Reference: www.puppet.com/support-services/customer-support/support-plans

SALTSTACK

SaltStack Enterprise : $15,000

Reference: www.saltstack.com/enterprise/

Ansible Enterprise Tower : $10,000

Reference: www.ansible.com/pricing

ANSIBLE

Chef Enterprise : $7,200

Reference: www.chef.io/pricing/

CHEF



Ansible Vs Puppet Vs Chef Vs SaltStack

- Conf. Lang.
- GitHub Activity
- Enterprise Cost
- Popularity
- Success Story

puppet

Contributors: **355**
Commits: **19,595**
Branches: **9**
Releases: **291**

SALTSTACK

Contributors: **1,041**
Commits: **49,193**
Branches: **11**
Releases: **82**

Contributors: **1,003**
Commits: **13,527**
Branches: **33**
Releases: **57**

ANSIBLE

Contributors: **369**
Commits: **12,089**
Branches: **177**
Releases: **231**

CHEF

Reference: www.slideshare.net/DanielKrook/caps-whats-best-for-deploying-and-managing-openstack-chef-vs-ansible-vs-puppet-vs-salt



Ansible Vs Puppet Vs Chef Vs SaltStack

- Conf. Lang.
- GitHub Activity
- Enterprise Cost
- Popularity
- Success Story

puppet

Configuration : **DSL** (Puppet DSL)
Not easy to learn, **administrator** oriented

SALTSTACK

Configuration : **YAML (Python)**
Easy to learn, **administrator** oriented
Python is **inbuilt** into most Unix and Linux deployments nowadays, so setting the tool up and running is quicker

Configuration : **YAML (Python)**
Easy to learn, **administrator** oriented
Python is **inbuilt** into most Unix and Linux deployments nowadays, so setting the tool up and running is quicker

ANSIBLE

Configuration : **DSL** (Ruby)
Steep Learning Curve, **developer** oriented

CHEF

- **DSL** : Domain Specific Language
- **YAML** : Yet Another Markup Language



Ansible Vs Puppet Vs Chef Vs SaltStack

Scorecard	Scalability (20%)	Setup (20%)	Availability (20%)	Management (20%)	Interoperability (20%)	Overall Score (100%)
Puppet	9.0	8.0	9.0	8.0	9.0	**8.6**
SaltStack	9.0	8.0	9.0	9.0	9.0	**8.8**
Chef	9.0	8.0	9.0	8.0	9.0	**8.6**
Ansible	9.0	9.0	9.0	9.0	9.0	**9.0**

- ✓ We have given equal weightage for all the parameters to calculate overall score
- ✓ You may calculate the overall score by changing the weightage of parameters depending on your requirement



Ansible Vs Puppet Vs Chef Vs SaltStack

Scorecard	Scalability (20%)	Setup (20%)	Availability (20%)	Management (20%)	Interoperability (20%)	Overall Score (100%)
Puppet	9.0	8.0	9.0	8.0	9.0	**8.6**
SaltStack	9.0	8.0	9.0	9.0	9.0	**8.8**
Chef	9.0	8.0	9.0	8.0	9.0	**8.6**
Ansible	9.0	9.0	9.0	9.0	9.0	**9.0**

- ✓ We have given equal weightage for all the parameters to calculate overall score
- ✓ You may calculate the overall score by changing the weightage of parameters depending on your requirement



Ansible Vs Puppet Vs Chef Vs SaltStack

- Scalability
- Ease of Setup
- Availability
- Management
- **Interoperability**

puppet

Puppet Master works only on Linux/Unix but Puppet Agent also works on windows

SALTSTACK

Salt Master works only on Linux/Unix but Salt minions can work on windows as well

ANSIBLE

Ansible supports windows machines as well but the Ansible server has to be in Linux/Unix machine

CHEF

Chef Server works only in Linux/Unix but Chef Client and Workstation can be on windows as well



Ansible Vs Puppet Vs Chef Vs SaltStack

- Scalability
- Ease of Setup
- Availability
- **Management**
- Interoperability

puppet

Not very easy to learn to manage the configurations as it uses its own language called Puppet DSL. Client pulls the configurations from the Server
Non-Immediate remote execution

SALTSTACK

Easy to learn to manage the configurations
Server pushes configurations to all the clients
Immediate remote execution

ANSIBLE

Easy to learn to manage the configurations
Server pushes configurations to all the nodes
Good for real-time application
Immediate remote execution

CHEF

You need to be a programmer to manage the configurations as it offers configurations in Ruby DSL. Client pulls the configurations from the Server

Ansible Vs Puppet Vs Chef Vs SaltStack

- Scalability
- Ease of Setup
- **Availability**
- Management
- Interoperability

puppet

Highly Available

It has **multi-master architecture**, if the **active master** goes down, the other master takes the active master place

SALTSTACK

Highly Available

It can have **multiple masters** configured, if one master is down, agents connect with the other master in the list

ANSIBLE

Highly Available

Runs with a single active node, called the **Primary instance**, if primary goes down, there is **Secondary instance** to take its place

CHEF

Highly Available

When there is failure in the **primary server** i.e. **chef sever**, it has a **backup server** to take the place of primary server

Ansible Vs Puppet Vs Chef Vs SaltStack

- Scalability
- **Ease of Setup**
- Availability
- Management
- Interoperability

puppet

Master-Agent

Puppet server runs on master machine and **Puppet clients** runs as agent on each client machine

SALTSTACK

Master-Agent

Here Server is called as **master** and clients are called as **minions** which run as agents in client machine

Master-Node

It has only **master** running on server machine, but **no agents** running on client machine, uses **ssh** to login to client systems

Client machine VM requires no special setup, hence it is faster to setup

ANSIBLE

Master-Agent

Chef server runs on master machine and **Chef clients** runs as agent on each client machine

CHEF



Ansible Vs Puppet Vs Chef Vs SaltStack

- **Scalability**
- Ease of Setup
- Availability
- Management
- Interoperability

puppet

Highly Scalable

SALTSTACK

Highly Scalable

Highly Scalable

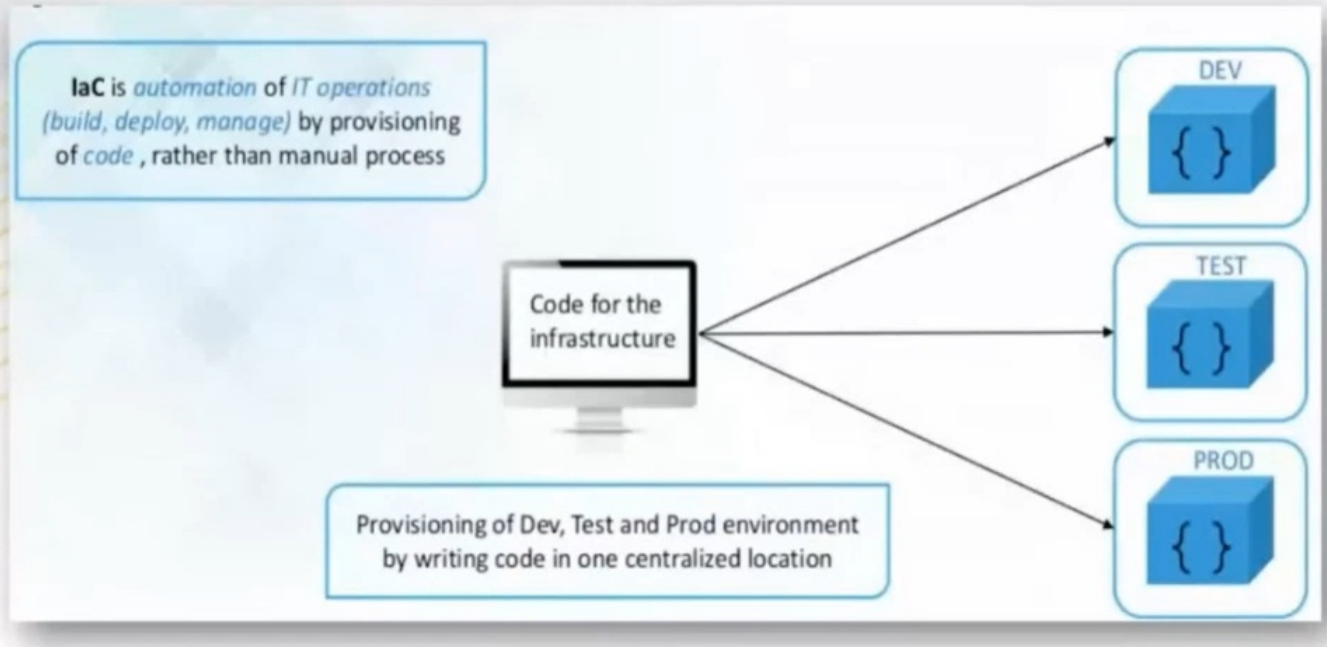
Highly Scalable

ANSIBLE

CHEF

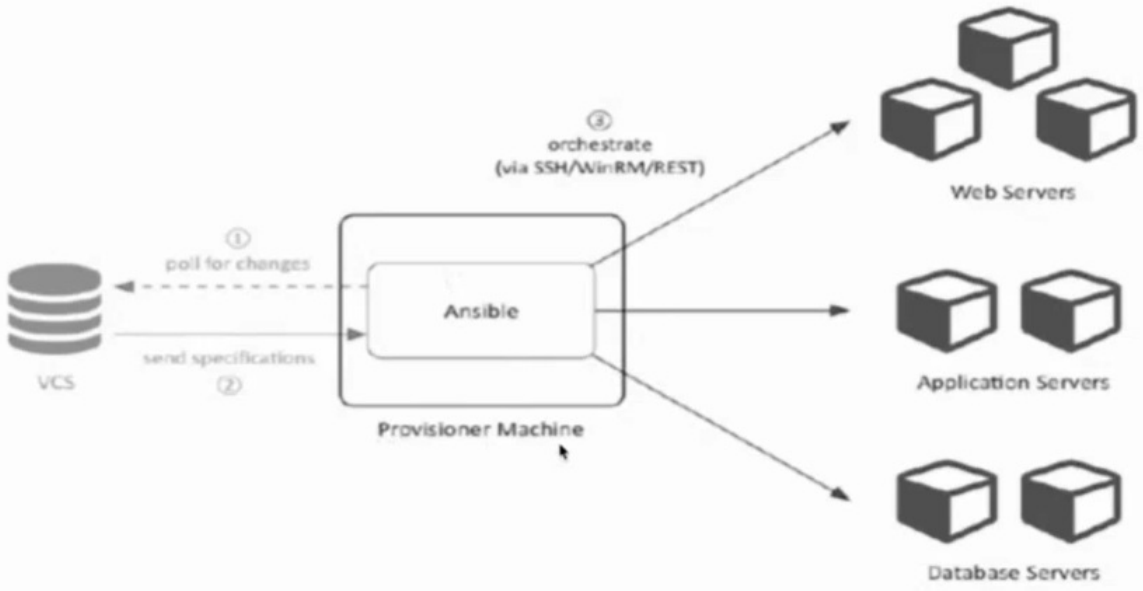


IaC

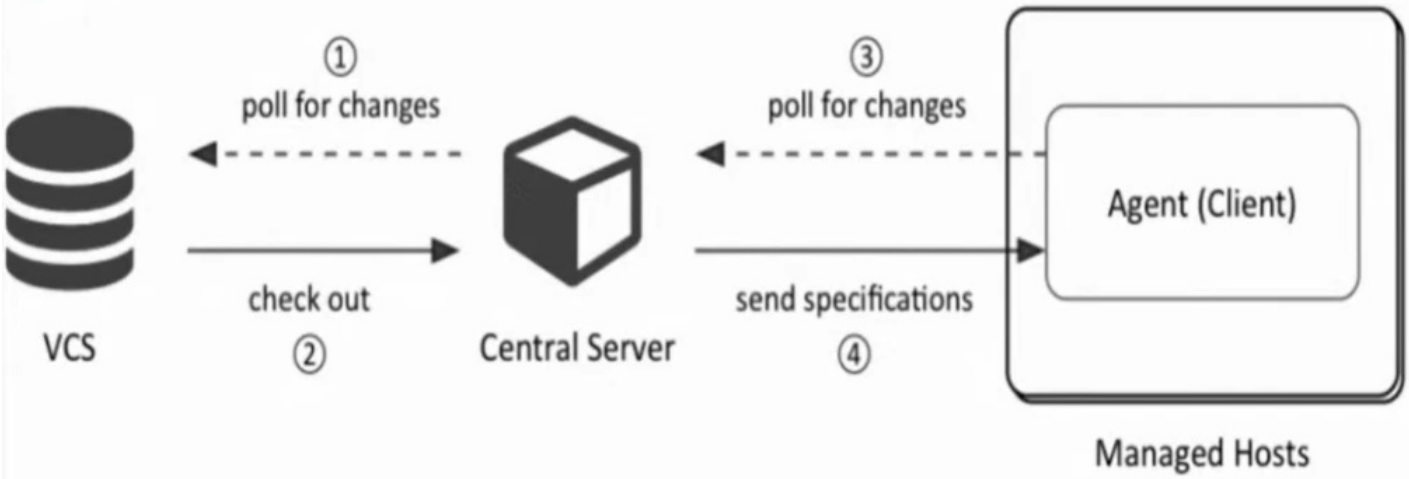


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Ansible's Agentless Architecture



Agent-Based Architectures



Advantages of Using Ansible

Agent-Based Systems	Agentless Systems
These systems need an agent and its dependencies installed.	No specific agent or third-party dependencies are installed.
Need to invoke the agent to run the configuration management tool. (as a service or cron job).	Invoke the run remotely
Parallel agent runs might be slow if they all hit the same server and the server cannot process several concurrent connections effectively.	Parallel agent runs might be faster than when all agents are contacting the same machine, but they might be constrained by the number of SSH connections.
The agent's installation and permissions need to be taken care of along with the configuration of the agent itself.	Remote connections can log in as a specific user and with the right level of user support since it's SSH-based.

Advantages of Using Ansible



Simple



Agentless



Powerful & Flexible



Efficient

What Ansible can do?



Provisioning



Configuration Management



Orchestration



Application Deployment



Security & Compliance

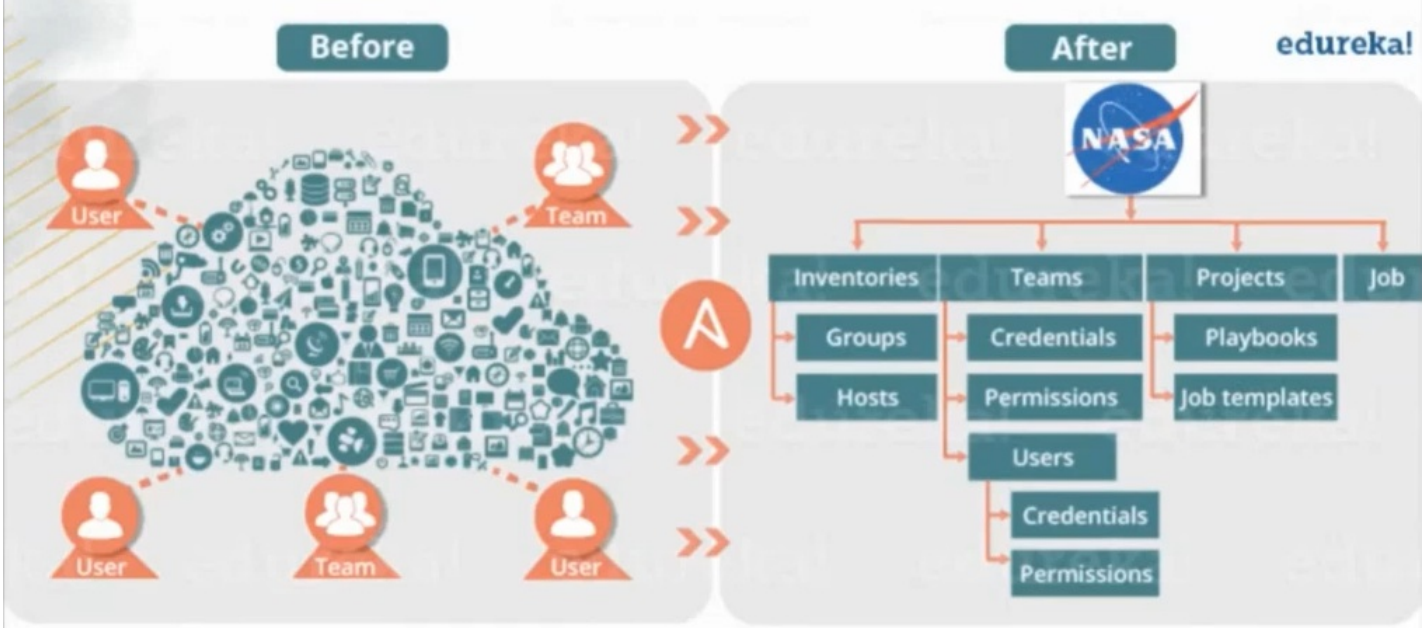
Need of Ansible ...





Due to test/production environment difference, few applications work on test but not on production. There should be a tool to handle the environment differences.

Need of Ansible ...



Need of Ansible ...

Deploy the application on 5 VMs





Deploy the application on 500 VMs





It is not possible for a human to deploy application on 500 VMs quickly, We need a tool that can do this job.



Need of Ansible ...

Before Ansible:

- ✓ system administrators do the following by hand:
  - ✓ Managing servers
  - ✓ Deploying & Installing software
  - ✓ Changing Configurations
  - ✓ Administrating Services

- ✓ Data Centers are growing
- ✓ Applications are becoming more complex
- ✓ Developers Release Applications frequently



ANSIBLE



Need of Ansible ...

We're seriously overloaded...!

- ✓ Building VM templates
- ✓ Configuring clusters
- ✓ Storage management
- ✓ Networking
- ✓ Software deployment
- ✓ User/Group Authentication/Authorization
- ✓ Performance Testing
- ✓ OS Upgrades
- ✓ Software Upgrades
- ✓ Development Labs
- ✓ Alerting/Monitoring
- ✓ Customer Support
- ✓ Troubleshooting
- ✓ Security Patching



Need of Ansible ...

- ✓ Many developers and system administrators **manage servers** by logging into them via SSH, making changes, and logging off. Some of these changes would be documented, some would not !
- ✓ If an admin needed to make the same **change** to many servers, the admin would manually log into each server and repeatedly make this change !
- ✓ If the admins wanted to set up a new server exactly like one that is currently running, they would need to spend a lot of unnecessary time to reinstall, update, and tweak everything to get the new server to run.
- ✓ And how about the new stranger employees ?!



Exercise 1: Install Ansible

1. Setup a new virtual machine based on Centos
2. Install Ansible on the new vm
3. Run a ping pong 🏓



Validate Ansible Installation

$ ansible -m ping localhost

```
localhost | SUCCESS => {
    "changed": false,
    "failed": false,
    "ping": "pong"
}
```




Validate Ansible Installation

$ Ansible --version

```
ansible 2.9.10
  config file = /etc/ansible/ansible.cfg
  configured module search path = [u'/usr/share/my_modules']
  ansible python module location = /usr/lib/python2.7/site-packages/ansible
  executable location = /usr/bin/ansible
  python version = 2.7.5 (default, Oct 30 2018, 23:45:53) [GCC 4.8.5 20150623
(Red Hat 4.8.5-36)]
```


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Install Ansible

Via apt:

```
$ sudo apt update
$ sudo apt install software-properties-common
$ sudo apt-add-repository --yes --update ppa:ansible/ansible
$ sudo apt install ansible
```




Install Ansible

Via Zypp:

```
$ zypper add repo https://download.opensuse.org/repositories/
systemsmanagement/SLE_15_SP1/systemsmanagement.repo
$ zypper refresh
$ zypper install ansible
```

Via pip:

```
$ sudo easy_install pip
$ sudo pip install ansible
```




Install Ansible

from source:

```
$ git clone git://github.com/ansible/ansible.git
$ cd ansible/
$ source ./hacking/env-setup
```

Via Yum:

```
$ yum install epel-release
$ yum install ansible
```




نصب سرور Ansible برای ویندوز نیست!

Ansible Install

Linux Base



Use Ansible on any



UNIX

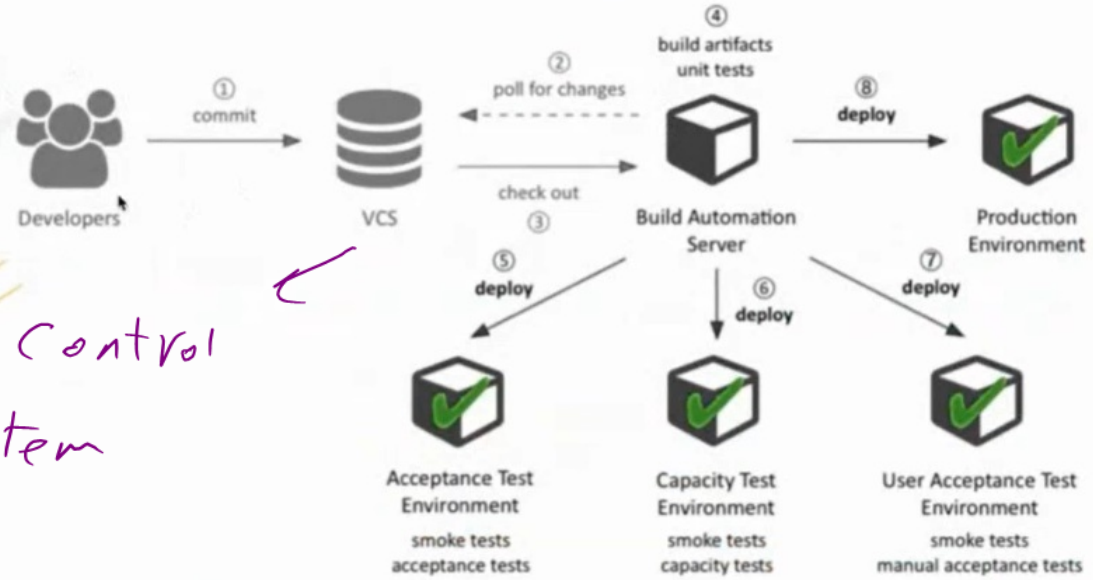
Ansible doesn't require any database and doesn't need any daemons running. ⇨ easier to maintain the Ansible versions and upgrade without any breaks

- ✓ Ansible can be run from any machine with Python 2 (version 2.7) or Python 3 (versions 3.5 and higher) installed. This includes Red Hat, Debian, CentOS, macOS, any of the BSDs, and so on.
- ✓ Windows is not supported for the control node.



Automation Deployment (PipeLine)

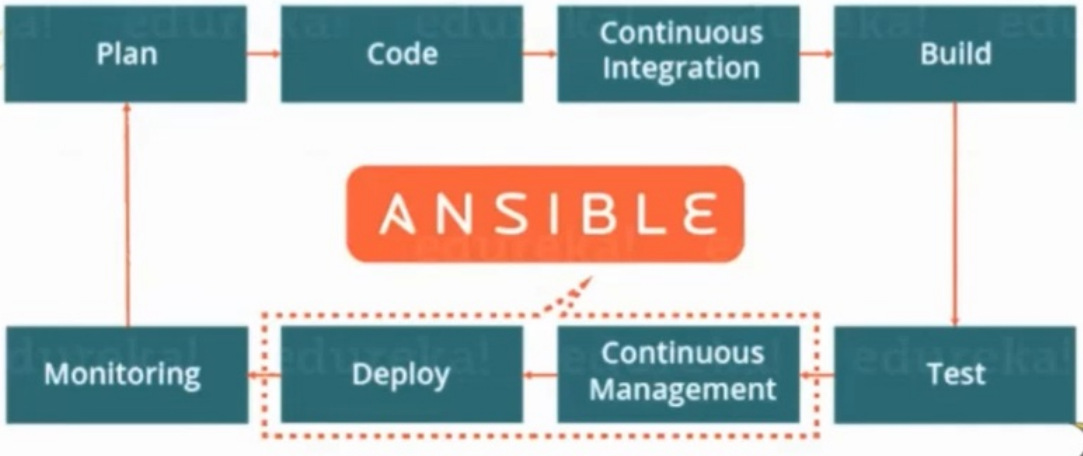
یک سرور خام در اختیار گرفتیم و بیاد همه مراحل تستینگ OS ها انجام بدن



Version Control System

What is Ansible?

- ✓ Ansible is an open source IT Configuration Management, Deployment & Orchestration tool.
- ✓ Ansible describes how a group of machines should be configured or what actions should be taken on them.



Learn Ansible In Iran Linux House

Automation for everyone



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Automation Deployment (Why?)

- Create application runtime environments **on demand**
- **Fast**, **reliable**, **repeatable** and **predictable** outcomes
- **Consistent** environments in staging and production
- Establish **fast feedback loops** you can react upon
- Makes release days **riskless**, almost boring



The benefits of CI/CD

- ✓ Easy to debug and change
- ✓ Release and delivery speed increases
- ✓ Increased code quality
- ✓ Reduces the cost
- ✓ Increased flexibility

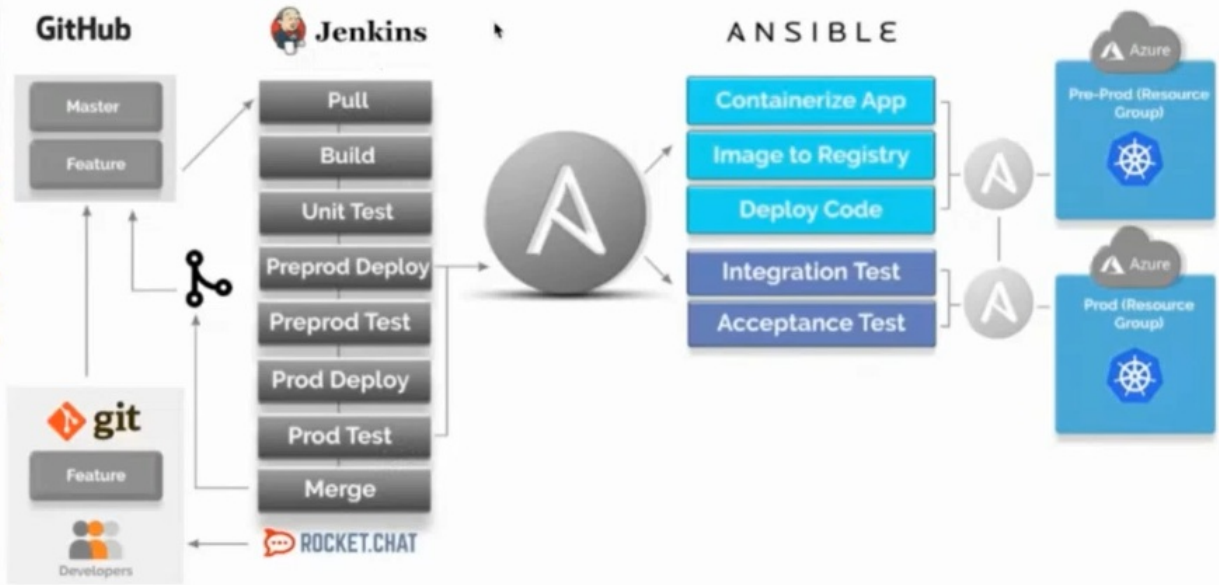


Why Automation?

- Tasks in code
- Collaboration
- Eliminate errors
- Write once
- Laziness
- …

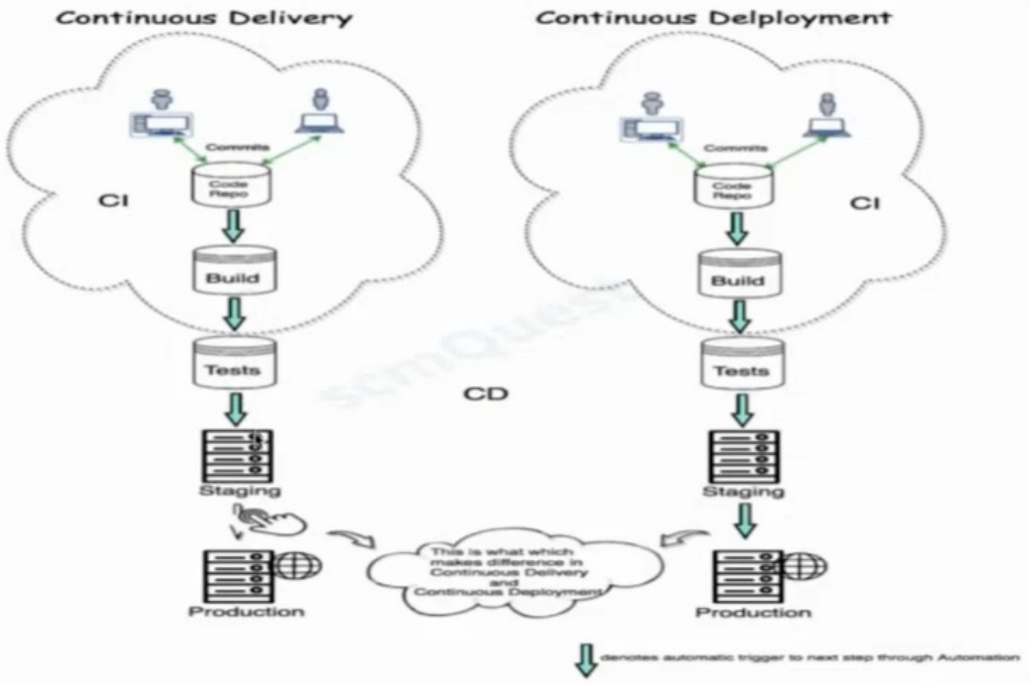


Automation Deployment (Why?)

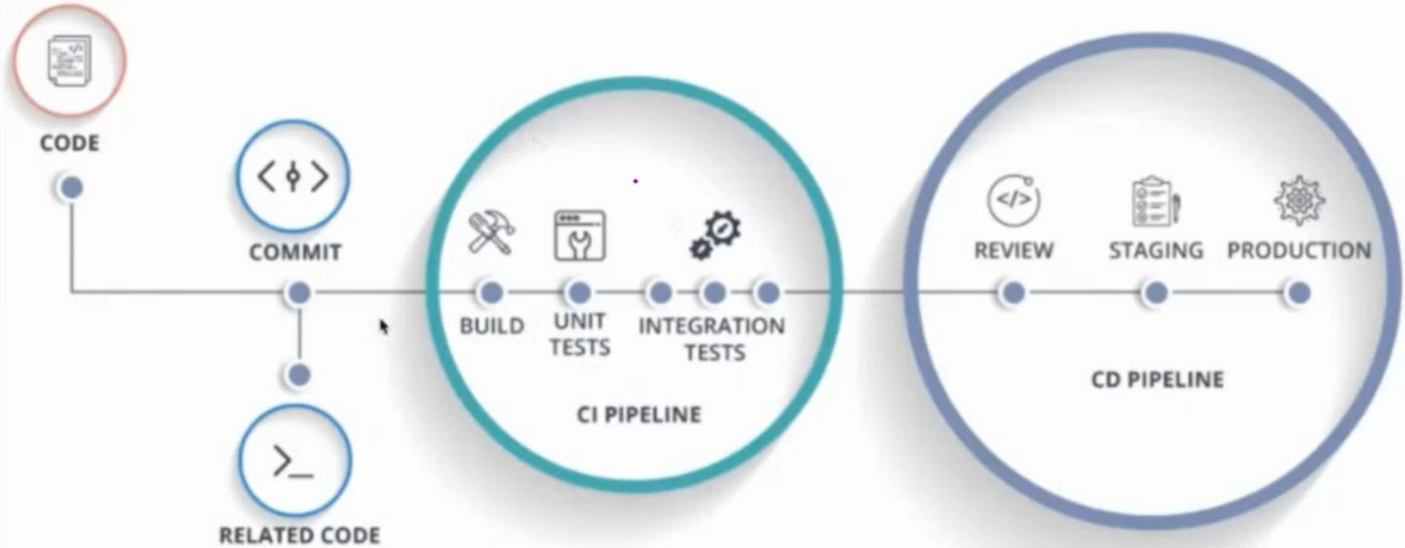




Continuous Integration vs Continuous Delivery vs Continuous Deployment



Continuous Integration vs Continuous Delivery



Automation Deployment (What?)

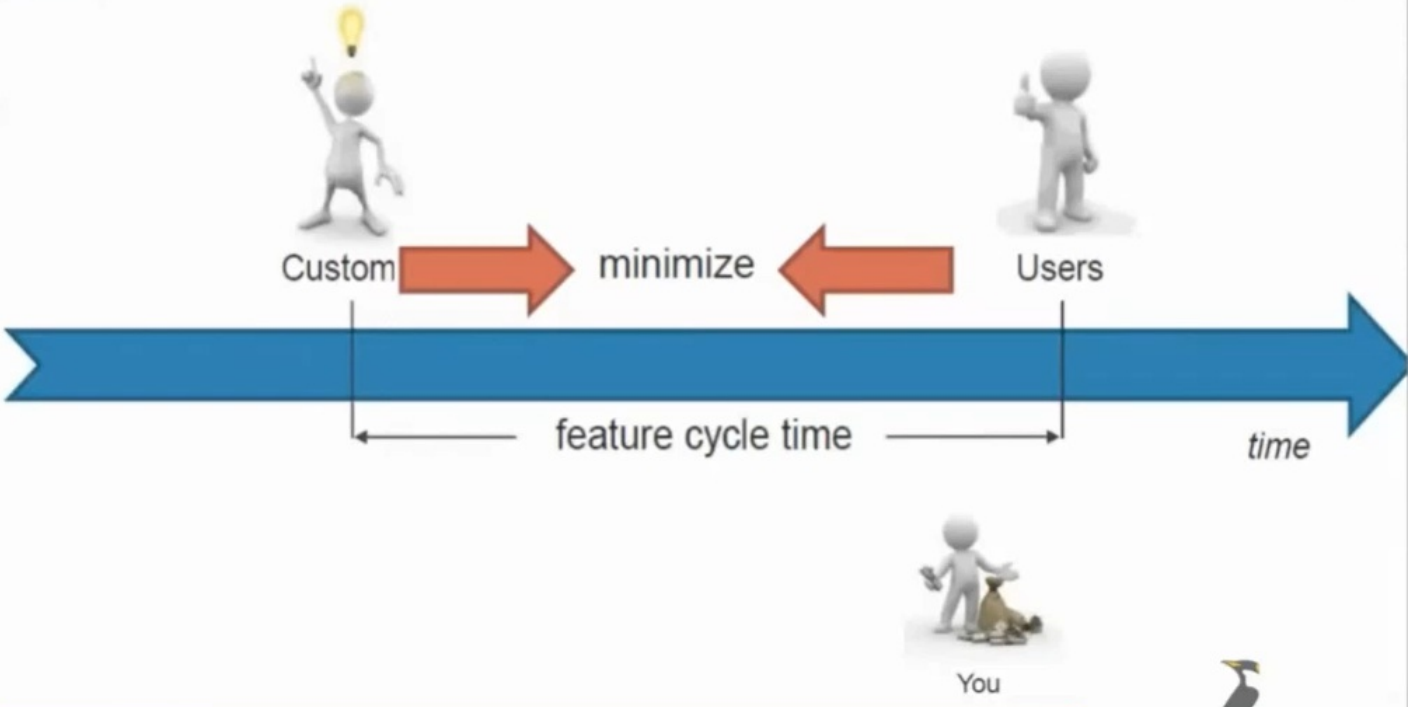
- Operating Systems, Drivers
- Middleware, Databases, etc.
- Applications, Dependencies, Data

} + Configuration

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Automation Deployment (Why?)



The Primary Goal of Continues Delivery

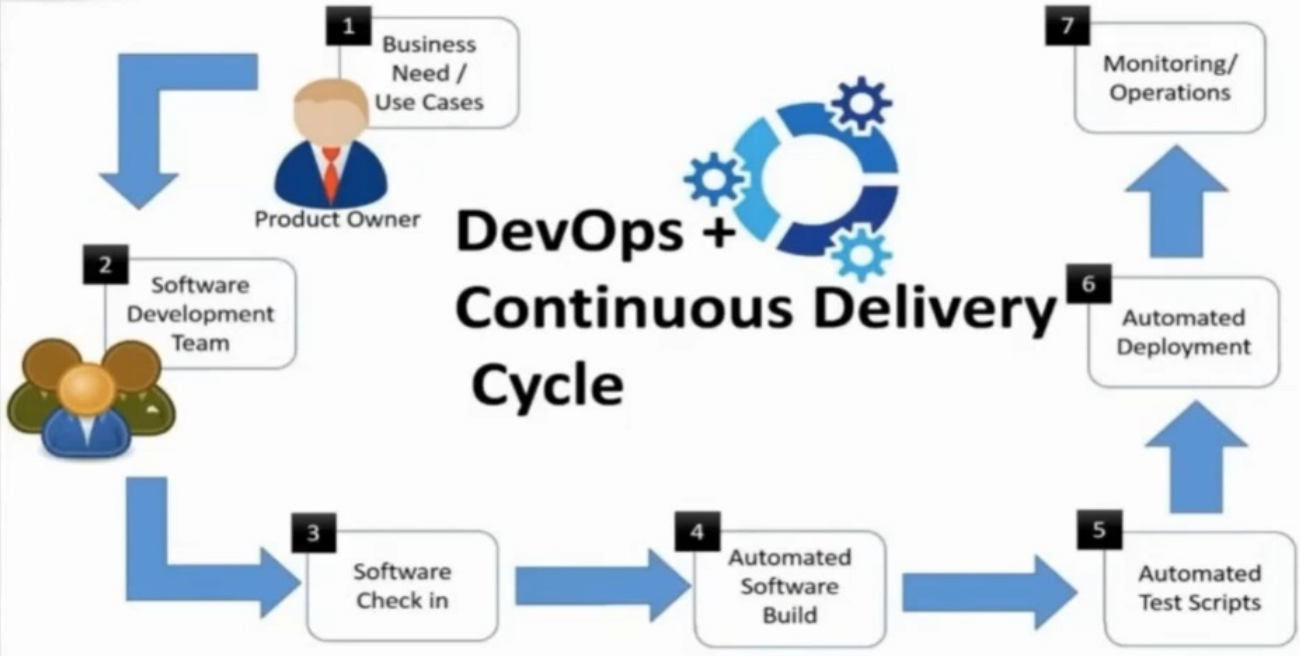
Cycle Time Reduction

هدف CD؟ کاهش زمان!



- The product cost: $1 billion
  The Development cost: 10% of revenue for a company
- To improve the efficiency:
  - Option 1: 10% ↓ in development cost ➡ $10 million reduction in development cost
  - Option 2: 10% ↓ in product development cycle ➡ $100 million add to top line revenue (1.2 months earlier)

Continues Delivery



Continues Delivery ...



- Continuous Delivery is about putting the release schedule in the hands of the business, not in the hands of IT.
- Making sure your software is always production ready throughout its entire lifecycle - that any build could potentially be released to users.

Continuous Integration benefits

برای مشخص شدن باگ ها تو تست ها و فیکس کردن

- If you are going to fail, then fail early
- Increases confidence in the software
- Team Communication

ارتباطات تیمی !

- Risk Mitigation

پایین آمدن ریسک توسعه نیری

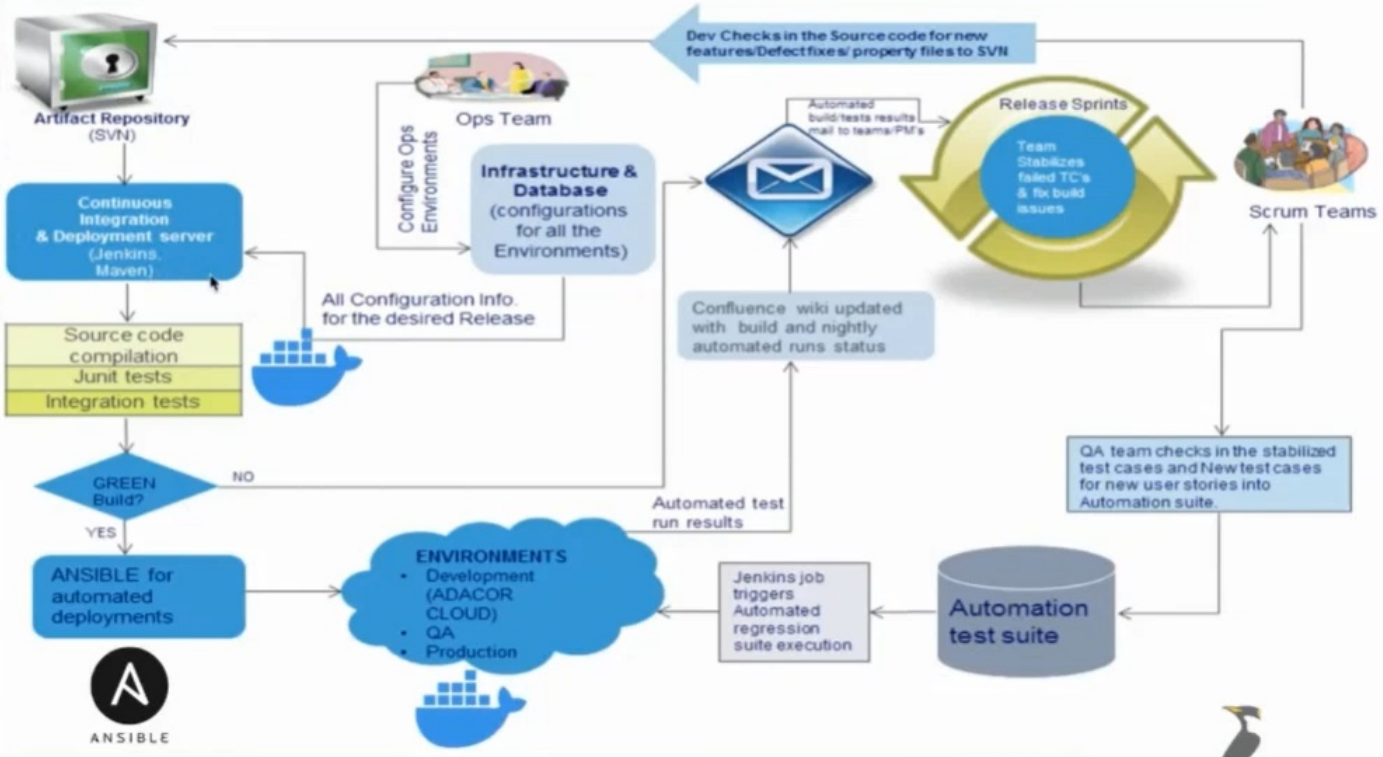
Continuous Integration

که بتونید یکپارچه برای dev ها قرار بدیم که بدونه بصورت خودکار تست رو بنویسه - تست کنه

CI is a software development practice which frequently refers to integrating, building and testing code within the development environment.

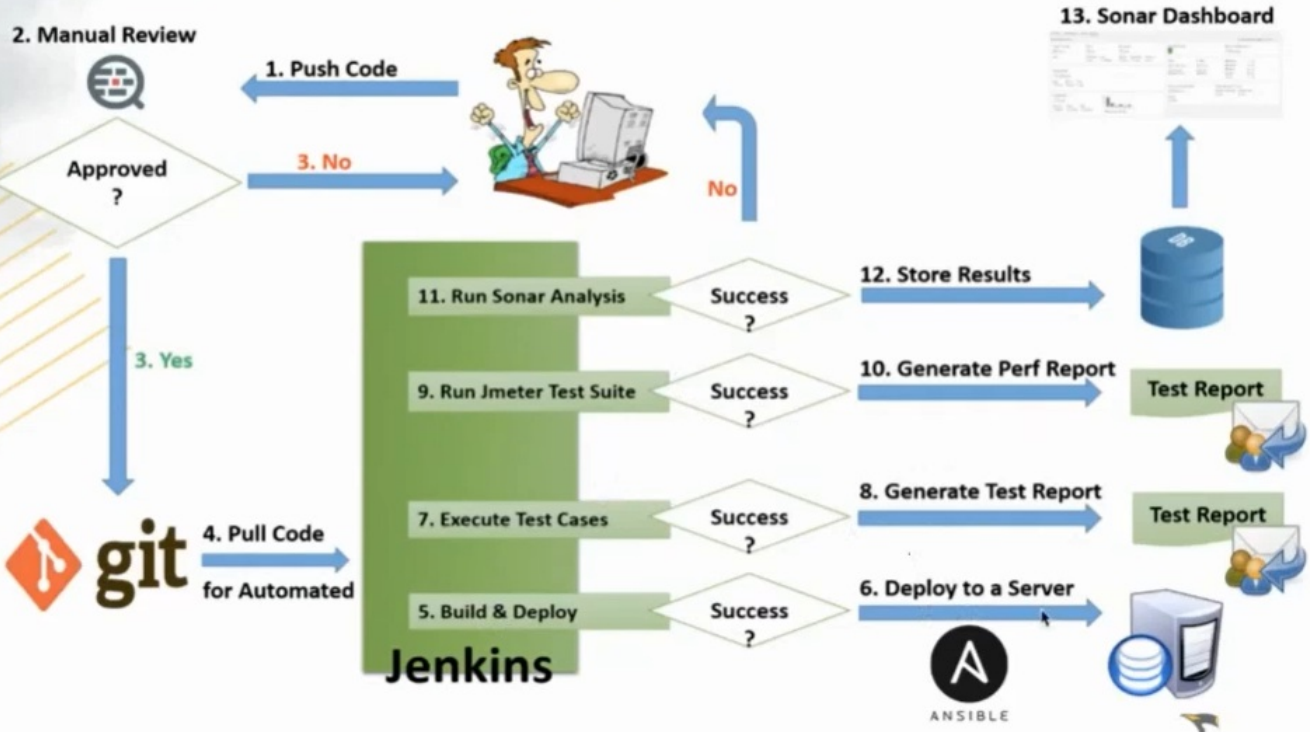


Automation Deployment

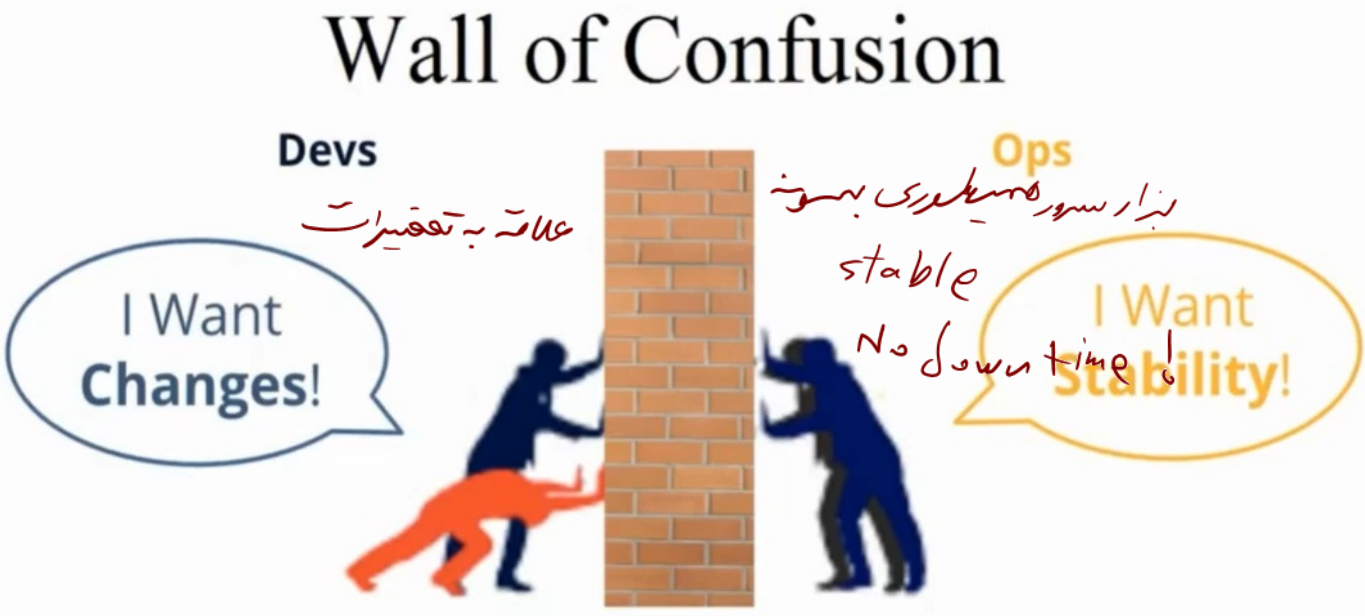


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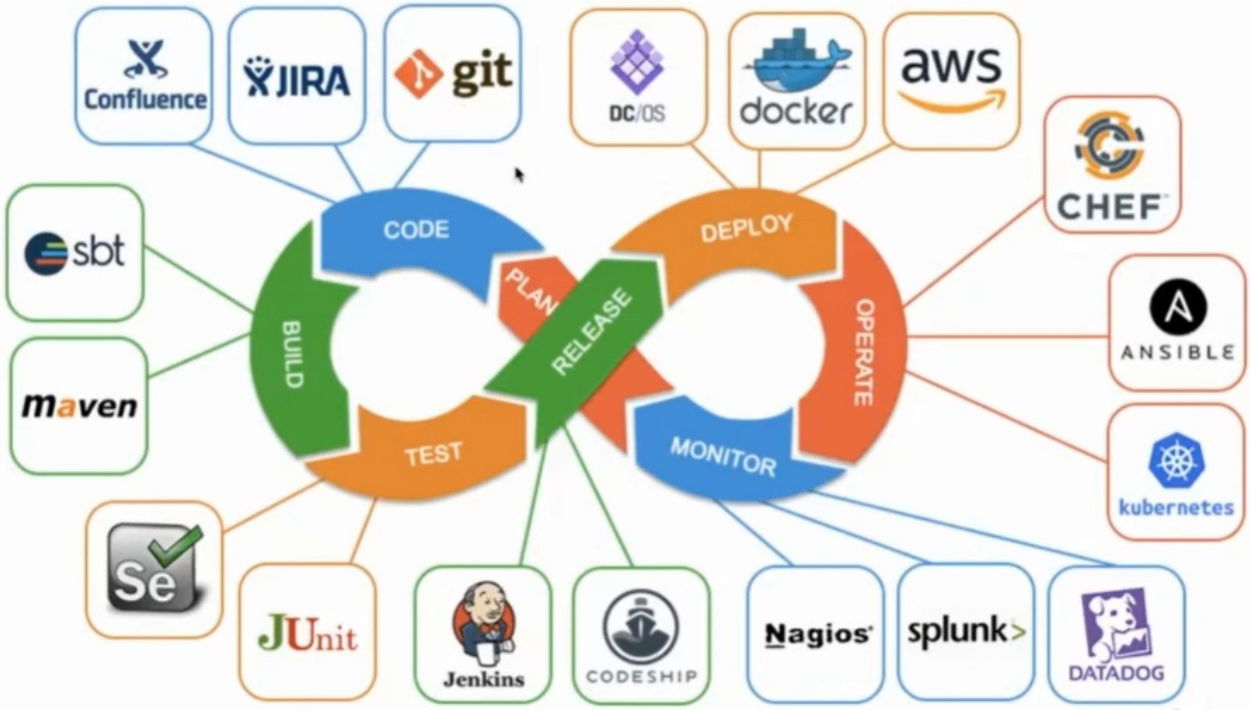
DevOps Automation (Continue)



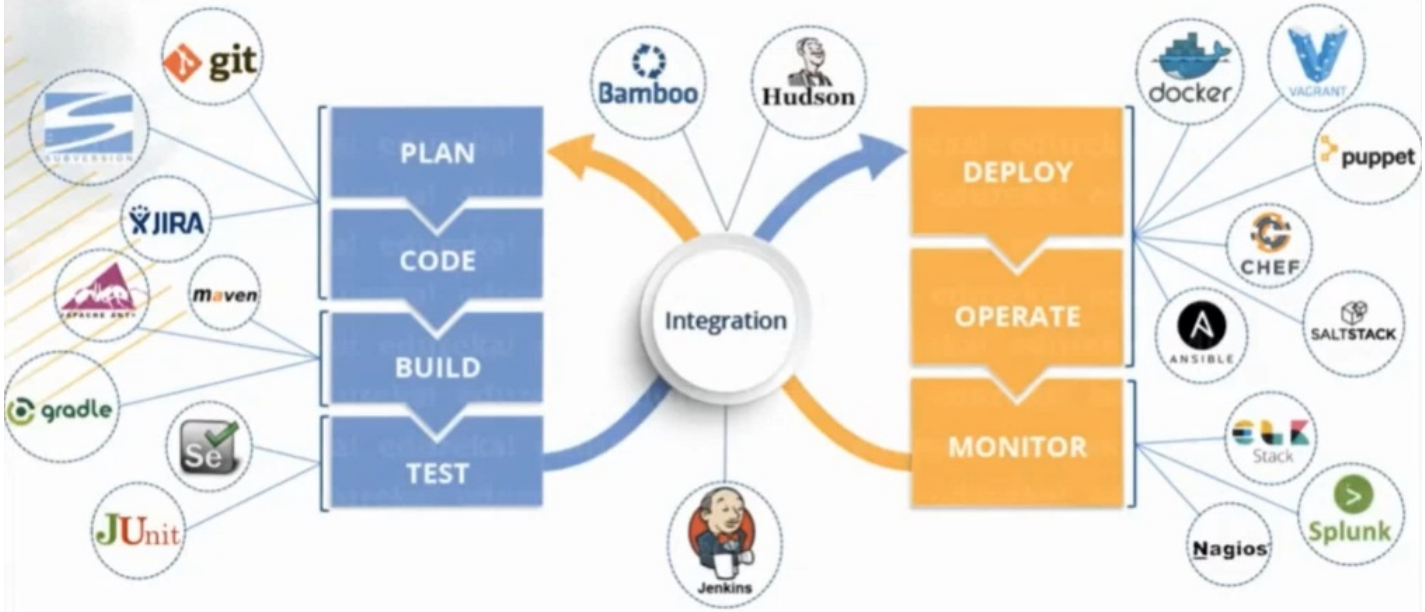
DevOps: Couse of Tension



DevOps Automation



DevOps Automation ...



DevOps

یک فرهنگ است

DevOps is a **culture**, movement or practice that emphasizes the **collaboration** and **communication** of both software developers and other IT professionals while **automating the process of software delivery and infrastructure changes**.



ما در DevOps بیشتر اتومات کردن می‌خوایم

مگر اینکه شرکت نگذارد

DevOps is a **cultural** and operational model that fosters **collaboration** to enable high-performance IT to achieve **business goals**.

Learn Ansible In Iran Linux House
Title: DevOps

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- docker_swarm module
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- AWX - hosts, groups and inventory
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- AWX - Schedule templates, notification and permissions

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Ansible

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